

IoRT Data Collection With LEO Satellite-Assisted and Cache-Enabled UAV: A Deep Reinforcement Learning Approach

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Abstract—Space air ground integrated network (SAGIN), leveraging low earth orbit (LEO) satellites and Unmanned Aerial Vehicles (UAVs), is expected to play a key role in providing services to Internet of Remote Things (IoRT) in the sixth generation (6G) communications. Our considered SAGIN incorporates a cache node on the UAV to cope with the data rate fluctuation in the backhaul link (UAV to satellite), allowing temporary storage of collected data during low data rate periods. In this paper, we aim to minimize the completion time of data collection in SAGIN by optimizing the UAV trajectory, IoRT device association scheme, and data caching policy (whether to store data temporarily or not in the UAV). Since the formulated problem is challenging to solve by using traditional optimization methods due to the unknown number of decision variables and the changing environment, we propose a deep reinforcement learning (DRL)-based algorithm to efficiently solve it. Simulation results demonstrate that our proposed algorithm requires less time to complete data collection compared to both the circular trajectory scheme and the no-cache node scheme under various setups. Moreover, our proposed algorithm can adapt to uneven data distribution by approaching closer to the IoRT nodes with large data sizes, and it can also mitigate the influence of backhaul link fluctuations with the aid of the cache node.

Index Terms—Low Earth orbit (LEO), unmanned aerial vehicle (UAV), deep reinforcement learning (DRL), trajectory optimization.

I. INTRODUCTION

BEYOND fifth generation (5G) or sixth generation (6G) communications [1], [2], which aim to provide global coverage and ubiquitous service to users, have attracted growing attention from both the industry and academia. Internet of Remote

Things (IoRT) has become one of the most important application scenarios in the next generation communication [3], [4] due to its role in remote surveillance such as animal protection, forest conservation, natural disaster and climate change monitoring, etc. IoRT is a subset of IoT and focuses on connecting and enabling communication with devices and sensors in remote or depopulated areas, such as forests, oceans, and deserts. However, IoRT poses challenges to the existing network architecture since IoRT devices are usually deployed in depopulated areas which lack terrestrial base stations.

The fast development and widespread deployment of Low Earth Orbit (LEO) satellites in recent years (e.g., Starlink, Kuiper, and OneWeb, etc.) present a solution [5], [6], [7] to overcome the above challenge. These satellites offer a means to provide service to IoRT devices in remote areas without the need for extensive ground infrastructure. Building ground infrastructure along the entire route would be inefficient when the goal is to connect devices at the end of the link. Additionally, irrespective of the high costs, constructing ground facilities may be infeasible for hazardous or hard-to-reach areas. LEO satellites are capable of reaching places where cables cannot, ensuring ubiquitous and continuous service. However, due to the large propagation loss and limited energy capacity of IoRT devices, directly offloading data of IoRT devices to the LEO satellites is often impractical, especially for large data volumes such as videos or high-definition pictures collection.

Unmanned Aerial Vehicle (UAV) [8], known for its flying maneuverability, deployment flexibility and low operational cost, has attracted growing interests and investigations in various applications, e.g., service recovery in disaster-stricken areas [9], performance improvement in existing cellular infrastructures [10], [11], and data collection for Internet of Things (IoT) [12]. Leveraging its high mobility and larger transmit power, UAV can build much better connections with both the ground IoRT devices (by approaching each of them in proper order) and LEO satellites (by transmitting with larger power). Therefore, we propose to utilize the UAV as a relay between the ground IoRT devices and the LEO satellites to expedite the data collection process. On the other hand, the rapid movement of LEO satellites and accompanying frequent handovers lead to substantial fluctuations in link capacity [13], [14], especially for users situated far from the equator in a Walker Delta constellation

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system [15] or service requests during the high network traffic periods. The data collection time can thus be increased if the UAV offloads the data during low rate periods of the backhaul link (UAV to satellites), even if the access link (IoRT devices to UAV) has a much higher data rate. To mitigate the influence of link capacity fluctuations, we further propose equipping the UAV with a cache node. This enables the temporary storage of collected data from IoRT devices in the UAV during low data rate periods of the backhaul link. Subsequently, data can be offloaded to the satellites once the link capacity improves. UAVs can be classified into two types: 1) the fixed wing UAV and 2) the rotary wing UAV. The rotary wing UAV can hover in a stationary point while the fixed wing needs to glide over the air to stay aloft. In addition, the fixed wing UAV is capable of carrying a heavier payload and has a longer flight endurance. Moreover, there is no requirement for the UAV to maintain a strictly stationary hovering position during the data collection process. Hence, we consider to deploy a fixed wing UAV to assist the data collection of IoRT devices. In the remaining sections of the paper, the term “UAV” will refer specifically to the fixed-wing UAV.

Our considered Space Air Ground Integrated Network (SAGIN), which operates within the framework of airspace and NTN (ASN) [16], consists of LEO satellites in the space, a cache-enabled UAV in the air and IoRT devices on the ground. By leveraging the global coverage provided by LEO satellites and the mobility of UAV, SAGIN enables data collection from remote and inaccessible areas that are typically challenging to reach using traditional ground-based approaches. Moreover, the cache node equipped on the UAV will mitigate the influence of data rate fluctuations in the backhaul link and further accelerates the data collection process. Despite the aforementioned advantages, there are still some challenges that need to be carefully handled to guarantee the performance of the considered SAGIN.

- 1) How to jointly optimize the trajectory of the UAV, the association scheme of IoRT devices and the caching policy of collected data? Trajectory design (or path planning) of the UAV has always been the core issue of UAV-assisted data collection. An optimized trajectory can potentially reduce the mission time and the energy consumption of the UAV. Furthermore, by associating with IoRT devices that have favorable channel conditions and properly adjusting the caching policy of the collected data based on the channel conditions of the backhaul link, the completion time can be further reduced. Note that these three problems are mutually dependent and need to be jointly considered to minimize the completion time of data collection. However, the joint optimization of these three variables introduces additional complexity, resulting in a highly complex non-convex and nonlinear problem formulation.
- 2) How to tackle the unknown number of decision variables? Note that the continuous time dimension of the data collection process is discretized into time intervals to simplify the complexity of the problem. Although this discretization into time slots facilitates analysis, the exact number of time slots required to complete the data collection remains unknown. Consequently, the number

of decision variables is also uncertain. This makes the formulated problem challenging to solve using traditional optimization methods.

- 3) How to make adaptive decisions in a time-varying environment? The fast movement of LEO satellites leads to time-varying channel conditions, which will, in turn, affect data collection in the access link. As a result, the data rate of the backhaul link cannot be considered as constant during the data collection process. In other words, the formulated problem is not a static optimization problem, as the decision variables or constraints may change over time.

Deep Reinforcement Learning (DRL) has demonstrated its efficacy in solving complex problems that are not easily tackled using conventional optimization techniques [17], [18], [19], [20]. Different from traditional optimization methods which assume static relationships between inputs and outputs, the learning-based methods can adapt and learn through interactions with the dynamic environments. The agent takes actions in the environment, receives feedback in the form of rewards or penalties, and learns to improve its decision-making over time [21]. Moreover, DRL can effectively approximate the optimal solution without exhaustively exploring every possible state. Instead, it generalizes from what it has learned to unseen or unvisited states to acquire a near-optimal solution, which guarantees the sample efficiency and convergence of the algorithm. By tuning the parameters of individual neurons in the neural network, the neural network can be used as function approximators to approximate the value function or policy function. This property enables neural networks to capture the underlying dynamics of the environment, which are often complex and non-linear. Additionally, neural networks can continuously learn and update their parameters based on new experiences, allowing them to adapt to changing environments and improve performance over time. The advantages of DRL make it an excellent solution for tackling the challenges posed by our considered SAGIN. Therefore, we reformulate the problem as a Markov Decision Process (MDP) and propose a DRL-based algorithm to learn the near optimal solution of the UAV trajectory, the IoRT devices' association scheme and the caching strategy of collected data. The main contributions of this article are listed as follows;

- 1) We consider the dynamics of the LEO satellite link capacity caused by the rapid movement of satellites. Rather than assuming a constant data rate for the backhaul link (which is impractical considering the time span of data collection), we consider its fluctuations and aim to obtain an adaptive solution that accounts for the time-varying environment.
- 2) We further propose to equip the UAV with a cache node to cope with data rate fluctuations in the satellite link. This cache node serves as a buffer between the access link and backhaul link. In other words, during low data rate periods, the collected data will be temporarily stored in the cache node to circumvent the limitations of the backhaul link. Note that the UAV to satellite data rate is determined by the slower of the two, the access link or the backhaul link.

Without cache node, the data collection process will be bottlenecked by the satellite link during low rate periods and will greatly increase the data collection time.

- 3) We consider a SAGIN architecture comprising LEO satellites in the space, cache-enabled UAV in the air, and IoRT devices on the ground. We aim to minimize the completion time of data collection without knowing the number of time slots (i.e., the number of decision variables is unknown).
- 4) We reformulate the original problem, which involves an unknown number of variables and a time-varying backhaul capacity, as an MDP and design a DRL-based algorithm to minimize the completion time of data collection. In addition, we assess the performance of the proposed DRL-based algorithm by considering various relevant parameters.

The remainder of this article is organized as follows. We present the related works in Section II. The system model of SAGIN and the completion time minimization problem are given in Section III. Section IV outlines our approach, where we reformulate the original problem as an MDP and propose a DRL-based algorithm to effectively address it. In Section V, simulation and numerical results are provided to evaluate the performance of the proposed DRL-based algorithm. Finally, the conclusion is given in Section VI.

II. RELATED WORKS

A. UAV Solely-Assisted Communications

UAV solely-assisted (without LEO satellites) communications has been widely studied. For example, Xu et al. [22] investigated the deployment of UAVs to assist data collection in a large-scale IoT network. They aimed to minimize the number of UAVs needed by designing the trajectories of the UAVs while ensuring the completion time of each UAV not to exceed a specified threshold. Chu et al. [23] studied the joint speed control and energy replenishment optimization in the context of UAV-assisted IoT data collection. Since the UAV operates under a dynamic and uncertain environment, an MDP is proposed to model the decision-making problem. Considering the limited computing capability and energy resource, a novel learning framework that allows UAVs to share and transfer learning information is developed to reduce the learning time. Cheng et al. [24] proposed to deploy a UAV at the edge of three adjacent cells to help offload traffic from Macro Base Stations (MBSs). They aimed to maximize the sum rate of all edge users served by the UAV by designing the trajectory of the UAV subject to each user's data rate requirement. The formulated mixed-integer nonconvex problem is then decomposed into two separate convex problems: the UAV trajectory design problem and the cell edge user scheduling problem. These two problems are solved alternately until a predefined threshold is reached. Liu and Zheng [25] studied a UAV-enabled environment monitoring heterogeneous network, where a rotary wing UAV is dispatched to help the MBS collect time-constrained data. They aimed to minimize the mission completion time by determining the trajectory and flying speed of the UAV, and the visiting sequence

of monitoring areas. To make the formulated problem more tractable, a Successive Convex Approximation (SCA)-based algorithm is proposed to decompose the primal problem into two subproblem, i.e., the UAV path planning problem and the UAV speed optimization problem. However, the above works focus solely on utilizing UAVs to enhance the performance of IoT and fail to provide service to remote areas with no access to terrestrial infrastructure.

B. UAV-LEO Integrated Network

The UAV-LEO integrated network, with the inclusion of LEO satellites, provides distinct advantages in terms of ubiquitous service and extensive coverage compared to UAV solely-assisted communications and therefore attracts increasing attention recently. Tran et al. [26] considered LEO satellite and UAV-assisted downlink communications for delivering content to ground users. The UAV is equipped with a cache which can pre-store popular contents during off-peak hours to alleviate congestion in the backhaul link during peak hours. They aimed to maximize the minimum data rate of all users by jointly determining the trajectory and resource allocation of the UAV and the cache content placement policy. Cheng et al. [27] investigated the computing offloading issue in remote areas and presented a SAGIN edge/cloud computing framework where the UAVs and the satellites provide access to the edge computing and the cloud computing, respectively. An optimization problem which minimizes the sum delay of all tasks by jointly determining the computing resource allocation and the task execution scheduling is formulated. To tackle the dynamics of the environment, the primal problem is reformulated as an MDP and a DRL-based algorithm is proposed to learn a near-optimal policy. Wang et al. [28] studied the data collection of IoRT in SAGIN where multiple UAVs are utilized as amplify-and-forward relays between the ground smart devices and the LEO satellites. They aimed to maximize the system throughput by optimizing the UAV trajectory, uplink power allocation and the association scheme of smart devices. However, they assumed the LEO satellite would not switch during the whole data collection process, which is unrealistic considering the frequent handovers in the backhaul link. Jia et al. [29] considered IoRT data collection with the assistance of LEO satellites and UAVs. Depending on the requirements of the data (i.e., delay-sensitive or delay-tolerant), the collected data will be relayed by the LEO satellites instantaneously or stored in the UAV and offloaded to the MBS at the destination point. Their approach is not applicable if there are no accessible MBSs or if the total amount of data to collected exceeds the UAV's storage capacity. In addition, the assumption that the backhaul capacity is always larger than that of the access link is impractical considering the significant fluctuations. Note that all the above works assume the link capacity from LEO satellite to UAV or to ground users to be unchanged during the considered time span, but this assumption is impractical in most application scenarios due to the rapid movement of LEO satellites. For example, a LEO satellite deployed at an altitude of 550 km has a linear velocity of approximately 7.9 km/s, resulting in a complete orbit cycle of approximately 90 minutes [30]. As a

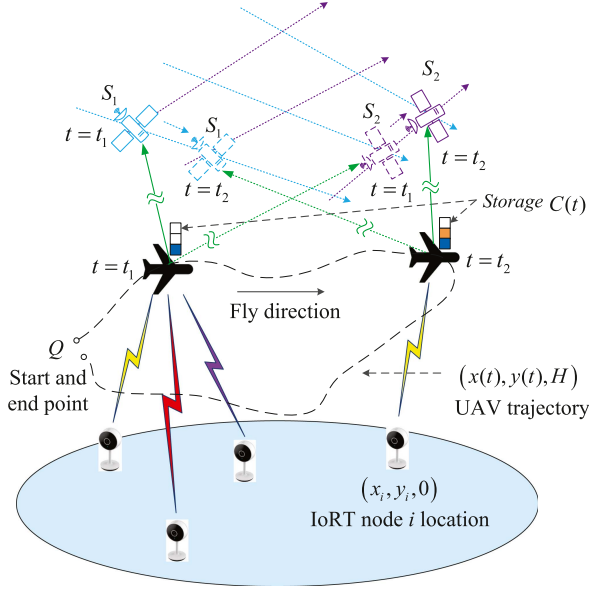


Fig. 1. Scenario of SAGIN.

result, the connected devices may experience significant channel fluctuations and may even lose connectivity to a particular LEO satellite after approximately 22.5 minutes (one quarter of the orbital period).

In contrast to the existing works, we study the employment of SAGIN for IoRT data collection with the consideration of fluctuations in the backhaul link. Specifically, we propose to equip the UAV with a cache node to temporarily store the collected data during low rate periods and offload the data to the LEO satellites during high rate periods. We further optimize the UAV trajectory, IoRT device association scheme and data caching policy to minimize the completion time of data collection.

III. SYSTEM MODEL AND PROBLEM FORMULATION

In Fig. 1, we illustrate the setup for uplink data collection of IoRT devices in a Space Air Ground Integrated Network. The IoRT devices are assumed to be deployed in a remote area with no accessible terrestrial infrastructure. A UAV is dispatched to assist in data collection from these IoRT devices and returns to the initial point for charging after the data collection is completed. The ground IoRT devices can choose to connect to the LEO satellites directly. However, due to the limited energy capacity of the IoRT devices and the long distance between the IoRT devices and the LEO satellite, the achievable data rate is very limited and cannot satisfy the demand of collecting a large amount of data (e.g., videos or high-resolution pictures). Moreover, the channel condition fluctuates drastically due to the fast movement of the LEO satellites and the resulting frequent handovers. The UAV serves as a relay or caching node, depending on the quality of the backhaul channel. By leveraging its proximity to the IoRT devices, higher transmission power, and higher flying altitude, the UAV improves the channel conditions and accelerates the data collection process. The architecture consists of LEO satellites in the space, one UAV in the air and a set of ground IoRT devices denoted by $\mathcal{I} = \{1, \dots, i, \dots, I\}$. At different time slots, the

UAV flies to different locations, and different IoRT devices may connect to the UAV. Similarly, LEO satellites change their positions over time. Consequently, the LEO satellite that can provide the best channel condition (shortest distance) to the UAV varies. In Fig. 1, the solid green line represents the LEO satellites with the best channel condition connected to the UAV, while the dashed green line represents LEO satellites in the UAV's neighbor list. As illustrated in Fig. 1, the UAV is connected to LEO satellite S_1 at time slot $t = t_1$ and then switches to LEO satellite S_2 at time slot $t = t_2$ when a better channel condition becomes available. The continuous time horizon is discretized into T time slots for simplicity. Each time slot $\delta(t)$ is assumed to be small enough such that the location of the UAV within each time slot is considered to be unchanged. We adopt orthogonal frequency division multiplex (OFDM) scheme when allocating the bandwidth resources in the access link. We further assume the flight altitude of the UAV to be fixed during the whole mission and denote the 3D coordinate of the UAV at time slot t as $q(t) = [x(t), y(t), H]$. Since we consider remote areas, such as deserts, oceans, and depopulated regions, which often feature natural terrain with few obstructing objects, the communication channels between IoRT devices and UAV are Line of Sight (LoS) link dominated [31], [32], [33]. Furthermore, the significantly higher operating altitude of the UAV (100 meters in this article) in comparison to ground base stations increases the probability of establishing LoS connections. Consequently, the Free Space pathloss model between IoRT device i and the UAV can be calculated as follows:

$$l_i(t) = \gamma_0 [d_i(t)]^2 \\ = \gamma_0 [(x(t) - x_i)^2 + (y(t) - y_i)^2 + H^2], \quad (1)$$

where γ_0 represents the pathloss value at the reference distance of 1 m, and $d_i(t)$ denotes the distance between the i -th IoRT device and the UAV at time slot t . $w_i = [x_i, y_i, 0]$ denotes the location of the i -th IoRT device and is assumed to be known *a priori* and fixed throughout the mission. Then, the maximum achievable rate from the i -th IoRT node to the UAV during time slot t can be calculated as

$$R_i(t) = b_i(t) \log_2 \left(1 + \frac{P}{l_i(t)\sigma^2} \right), \quad (2)$$

where P and $b_i(t)$ represent the transmit power and allocated bandwidth to IoRT i at time slot t , respectively. Note that the IoRT devices transmit data at a constant maximum allowable power. σ^2 denotes the environment noise power at the receiver. Similarly, the pathloss between the UAV and the LEO satellite can be calculated by [26], [29]

$$\xi^{U \rightarrow S}(t) = \beta_0 [d^{U \rightarrow S}(t)]^2 \\ = \beta_0 \{ [x^S(t) - x(t)]^2 + [y^S(t) - y(t)]^2 + [H^S - H]^2 \} \\ \stackrel{a}{\approx} \beta_0 \{ [x^S(t) - x(0)]^2 + [y^S(t) - y(0)]^2 + [H^S - H]^2 \} \\ \stackrel{b}{\approx} \beta_0 \{ [x^S(t) - x(0)]^2 + [y^S(t) - y(0)]^2 + (H^S)^2 \} \quad (3)$$

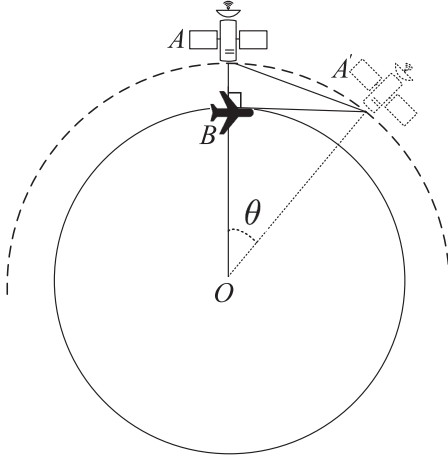


Fig. 2. LEO satellite orbit and UAV in the same plane.

where β_0 is the pathloss value between the LEO satellite and the UAV at the reference distance of 1 m, $d^{U \rightarrow S}(t)$ is the distance between the LEO satellite and the UAV at time slot t . $x^S(t)$, $y^S(t)$ and H^S are the longitude, latitude, and altitude of the LEO satellite at time slot t , respectively, which are known to the system operator. Step (a) in (3) holds because, compared to the movement distance of the satellite, the UAV can be considered stationary during the entire mission. Step (b) in (3) holds because the altitude of the UAV is negligible compared to that of the LEO satellite. The maximum achievable rate from the UAV to the LEO satellite during time slot t can be calculated as

$$R^S(t) = B^U \log_2 \left(1 + \frac{P^U}{\xi^{U \rightarrow S}(t) \sigma^2} \right), \quad (4)$$

where P^U and B^U are the transmission power and available bandwidth of the UAV, respectively. Since the connection with LEO satellite requires a clear sky view, which is also ensured by the high altitude of the UAV, a similar LoS channel-dominated model is used [27], [28], [29]. Because LEO satellite communication operates on a separate frequency bandwidth (e.g., Ku-Band and Ka-Band), the backhaul link does not suffer interference from the access link. All available bandwidth is allocated to the single UAV, which transmits at maximum power to expedite the data collection process.

Figs. 2 and 3 depict the minimum and maximum distances between the UAV and LEO satellite in scenarios where the LEO satellite orbit and the UAV are in the same plane and in difference planes, respectively. Note that A' represents the critical point for establishing a LoS connection between the UAV and the LEO satellite. Therefore, $\angle OBA' = 90^\circ$. A corresponds to the location where the LEO satellite is closest to the UAV, O is the center of the Earth, B denotes the location of the UAV. θ_1 and θ_2 in Fig. 3 represent the elevation angel of the LEO satellite and the latitude of the UAV, respectively. Then, we can easily derive

$$\min \{d^{U \rightarrow S}(t)\} = AB = H^S, \quad (5)$$

$$\max \{d^{U \rightarrow S}(t)\} = A'B = \sqrt{(r^E + H^S)^2 - (r^E)^2}, \quad (6)$$

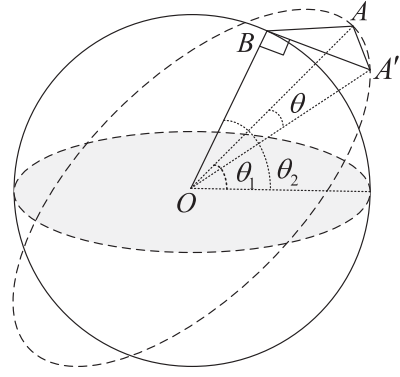


Fig. 3. LEO satellite orbit and UAV in different planes.

when the LEO satellite orbit and the UAV are in the same plane, where r^E is the radius of the Earth, and

$$\min \{d^{U \rightarrow S}(t)\} = \sqrt{(r^E)^2 + (r^E + H^S)^2 - 2(r^E)(r^E + H^S) \cos(\theta_2 - \theta_1)}, \quad (7)$$

$$\max \{d^{U \rightarrow S}(t)\} = \sqrt{(r^E + H^S)^2 - (r^E)^2}, \quad (8)$$

when the LEO satellite orbit and the UAV are in different planes. Depending on the quality of the backhaul channel, the UAV can choose to either directly relay the collected data to the LEO satellite or temporarily store it. In the relay mode, the backhaul capacity must be sufficient enough to accommodate the aggregated data rate in the access link, i.e.,

$$\sum_{i=1}^{|\mathcal{I}|} R_i(t) a_i(t) [1 - w_i(t)] \leq R^S(t), \forall t \in \mathcal{T}. \quad (9)$$

where $\mathcal{T} = \{t | 1 \leq t \leq T\}$ denotes the set of time slots, and $a_i(t)$ and $w_i(t)$ are the association indicator and caching indicator, respectively. Specifically, $a_i(t) = 1$ if IoRT device i is connected to the UAV at time slot t and $a_i(t) = 0$, otherwise. Similarly, $w_i(t) = 1$ implies that the collected data from IoRT device i is temporarily stored in the UAV at time slot t and $w_i(t) = 0$, otherwise. $|\mathcal{I}|$ denotes the cardinality of set $\mathcal{I}$, and its value is equal to the number of IoRT devices. In the caching mode, the amount of collected data in the upcoming time duration must be less than or equal to the remaining storage capacity of the UAV at the end of the previous time duration $C(t-1)$, i.e.,

$$\sum_{i=1}^{|\mathcal{I}|} R_i(t) \delta(t) a_i(t) w_i(t) \leq C(t-1), \forall t \in \mathcal{T}. \quad (10)$$

The mission is considered complete when all data from the IoRT devices has been successfully offloaded to the LEO satellites, indicating that there is no remaining data either in the IoRT devices or in the UAV. This condition can be expressed as

$$C(T) = C, \quad (11)$$

$$D_i(T) = 0, \forall i \in \mathcal{I}, \quad (12)$$

where C represents the capacity of the cache equipped on the UAV, $D_i(T)$ is the remaining data size of IoRT device i at time slot T . The total mission time can be calculated by summing up the time slots from the start of the mission until the completion of data offloading, which can be expressed as

$$\sum_{t=1}^T \delta(t). \quad (13)$$

Note that the value of T is unknown and needs to be determined in the problem formulation.

A. Problem Formulation

In this section, we tackle the challenge of designing the UAV trajectory, determining the association scheme of the IoRT devices, and formulating an efficient caching policy for the collected data in a network that leverages LEO satellite assistance and UAV-enabled IoRT. Our main goal is to minimize the completion time while considering various factors such as the need for data collection completion, the fluctuations in the backhaul link, the constraints of limited cache capacity and available bandwidth, and the maximum velocity allowed for the UAV. Based on the above discussions, the optimization problem is formulated as follows:

$$\mathbf{P0} : \min_{q(t), \delta(t), a_i(t), w_i(t), b_i(t), T} \sum_{t=1}^T \delta(t) \quad (14)$$

$$\text{s.t. } C(T) = C, \quad (15)$$

$$D_i(T) = 0, \forall i \in \mathcal{I}, \quad (16)$$

$$\sum_{i=1}^{|\mathcal{I}|} R_i(t) \delta(t) a_i(t) w_i(t) \leq C(t-1), \forall t \in \mathcal{T}, \quad (17)$$

$$\sum_{i=1}^{|\mathcal{I}|} R_i(t) a_i(t) [1 - w_i(t)] \leq R^S(t), \forall t \in \mathcal{T}, \quad (18)$$

$$q(0) = q(T) = Q, \quad (19)$$

$$\|q(t) - q(t-1)\| \leq V^m \delta(t), \forall t \in \mathcal{T}, \quad (20)$$

$$\sum_{i=1}^{|\mathcal{I}|} b_i(t) a_i(t) \leq B, \forall t \in \mathcal{T}, \quad (21)$$

$$b_i(t) \geq 0, \forall i \in \mathcal{I}, \forall t \in \mathcal{T}, \quad (22)$$

$$a_i(t), w_i(t) = \{0, 1\}, \forall i \in \mathcal{I}, \forall t \in \mathcal{T}, \quad (23)$$

where $\|\cdot\|$ stands for the Euclidean norm, Q represents the start point and end point of the UAV, V^m denotes the maximum allowed velocity of the UAV, and B represents the total available bandwidth. Constraint (19) implies that the UAV needs to return to the departure point after completing the data collection. The bandwidth resource budget of the access link is guaranteed by Constraint (21). Constraint (22) imposes bandwidth allocated to

IoRT devices to be non-negative. Constraint (23) imposes $a_i(t)$ and $w_i(t)$ to be binary variables.

Lemma 1: Given the trajectory of the UAV, the optimal association scheme of the IoRT devices is to associate the UAV with the IoRT device that has the best channel condition.

Proof: Suppose there are three IoRT devices associating with the UAV and transmitting data at certain time slot t . Devices 1, 2 and 3 have the best channel condition during time interval $\delta^1(t)$, $\delta^2(t)$ and $\delta^3(t)$, respectively. The duration of the time slot $\delta(t) = \delta^1(t) + \delta^2(t) + \delta^3(t)$. Next, we will prove that the amount of data collected in the multiple devices association scheme is less than that in the single device association scheme. First, the amount of data collected in the multiple devices association scheme D_{multi} can be expressed as

$$\begin{aligned} D_{\text{multi}} &= (b_1 \eta_1 + b_2 \eta_2 + b_3 \eta_3) \delta(t) \\ &= (b_1 \eta_1 + b_2 \eta_2 + b_3 \eta_3) (\delta^1(t) + \delta^2(t) + \delta^3(t)), \end{aligned} \quad (24)$$

where b_k and $\eta_k = \log_2(1 + P/[l_k(t)\sigma^2])$, $k = 1, 2$, and 3, represent the allocated bandwidth and spectral efficiency of device k , respectively. The above equation can be rewritten as

$$\begin{aligned} &(b_1 \eta_1 + b_2 \eta_2 + b_3 \eta_3) \delta^1(t) + (b_1 \eta_1 + b_2 \eta_2 + b_3 \eta_3) \delta^2(t) \\ &\quad + (b_1 \eta_1 + b_2 \eta_2 + b_3 \eta_3) \delta^3(t) \\ &\stackrel{c}{\leq} (b_1 \eta_1 + b_2 \eta_1 + b_3 \eta_1) \delta^1(t) + (b_1 \eta_2 + b_2 \eta_2 + b_3 \eta_2) \delta^2(t) \\ &\quad + (b_1 \eta_3 + b_2 \eta_3 + b_3 \eta_3) \delta^3(t) \\ &= (b_1 + b_2 + b_3) \eta_1 \delta^1(t) + (b_1 + b_2 + b_3) \eta_2 \delta^2(t) \\ &\quad + (b_1 + b_2 + b_3) \eta_3 \delta^3(t) \\ &\stackrel{d}{=} B \eta_1 \delta^1(t) + B \eta_2 \delta^2(t) + B \eta_3 \delta^3(t) \\ &= D_{\text{single}}. \end{aligned} \quad (25)$$

Step (c) in (25) holds since $l_k(t) \leq l_m(t)$, for $m \neq k$ during $\delta^k(t)$, i.e., $\eta_k \geq \eta_m$, for $m \neq k$ during $\delta^k(t)$. Step (d) in (25) holds because $B = b_1 + b_2 + b_3$ represent the total available bandwidth. (25) denotes the amount of data collected in the single device association scheme, which connects the UAV with only one IoRT device that has the best channel condition. In this case, all the available bandwidth is allocated to the only connected device. Hence, we have proved that the amount of data collected for the multiple devices association scheme is less than that of the single device association scheme. Note that the above proof process can be easily extended to scenarios where there are more than three IoRT devices associating with the UAV. ■

Based on Lemma 1, since there is only one connected IoRT device, $w_i(t)$ can be simplified to $w(t)$, and $b_i(t)$ can be omitted because all the available bandwidth will be allocated to the only connected device. We further assume equal time durations of all time slots, i.e., $\delta(1) = \delta(2) = \dots = \delta(T) = \delta$. Thus, Problem **P0** can be simplified as follows:

$$\mathbf{P1} : \min_{q(t), a_i(t), w(t), T} \sum_{t=1}^T \delta \quad (26)$$

$$\text{s.t. } \sum_{i=1}^{|Z|} R_i(t) \delta a_i(t) w(t) \leq C(t-1), \forall t \in \mathcal{T}, \quad (27)$$

$$\sum_{i=1}^{|Z|} R_i(t) a_i(t) [1 - w(t)] \leq R^S(t), \forall t \in \mathcal{T}, \quad (28)$$

$$\sum_{i=1}^{|Z|} a_i(t) = 1, \forall t \in \mathcal{T}, \quad (29)$$

$$a_i(t), w(t) = \{0, 1\}, \forall i \in \mathcal{I}, \forall t \in \mathcal{T}, \quad (15), (16), (19), (20) \quad (30)$$

Note that Constraint (21) is simplified to Constraint (29), and Constraint (22) is removed as $b_i(t)$ is omitted. However, solving **P1** remains challenging due to the indeterminate value of T , which results in an unknown number of decision variables. In addition, **P1** exhibits non-convexity due to the non-convex relationship between $R_i(t)$ and $q(t)$ [34]. Furthermore, the presence of integer variables $a_i(t)$ and $w(t)$ adds an additional layer of complexity to the problem. It is important to note that the fluctuation of the backhaul link capacity makes **P1** a dynamic problem, instead of a static one. This means that $R^S(t)$ cannot be treated as a constant value.

IV. PROPOSED DRL-BASED ALGORITHM

In this section, we reformulate **P1** as an MDP and propose a DRL-based algorithm to effectively solve it.

A. MDP Formulation

Note that **P1** is intractable and cannot be solved by using traditional optimization methods due to the factors mentioned above. However, MDP is well-suited for **P1** because the state in the current time slot depends only on the state of the preceding time slot and the action taken in that time slot. To apply the DRL-based algorithm, we need to reformulate **P1** as an MDP. The state space $\mathcal{S}$, the action space $\mathcal{A}$, and the reward function $\mathcal{R}$ for UAV trajectory design, association scheme of the IoRT devices and caching policy of the collected data can be defined as follows:

- 1) **State Space:** At the start of time slot t , the system state $\mathcal{S}_t$, which is inputted into the neural network to determine the subsequent action, comprises the current location of the UAV, the remaining storage of the cache, the data rate of the backhaul link and the remaining data quantity for each IoT node. Specifically,

$$\mathcal{S}_t = [q(t), C(t), R^S(t), D_1(t), D_2(t), \dots, D_{|Z|}(t)], \quad (31)$$

Therefore, the system state space can be expressed as

$$\mathcal{S} = \{\mathcal{S}_t | 0 \leq D_i(t) \leq D_i, \forall i \in \mathcal{I}, 1 \leq t \leq T\}. \quad (32)$$

- 2) **Action Space:** The action of the proposed SAGIN network $\mathcal{A}_t$ at time slot t determines the location of the UAV, the association scheme of the IoRT devices and the caching policy of the collected data at time slot $t+1$. Hence, $\mathcal{A}_t$

can be defined as

$$\mathcal{A}_t = [\phi(t+1), a_1(t+1), \dots, a_{|Z|}(t+1), w(t)], \quad (33)$$

where $\phi(t+1) = \{\text{east, west, south, north}\}$. To apply DRL, we have discretized the action space of UAV's trajectory into a set of limited available directions. Note that this approach has been widely adopted in previous works [35], [36] and is considered a good approximation when the step size is much smaller compared to the size of the moving area. Otherwise, to learn the optimal trajectories, the agent would need to numerically evaluate arbitrary UAV motion, a task that would require a much larger amount of computational resources. $a_i(t), w(t) = \{0, 1\}, \forall i \in \mathcal{I}, \forall t \in \mathcal{T}$. Since $\mathcal{A}_t, \forall t \in \mathcal{T}$ share the same available actions, the system action space can be defined as

$$\mathcal{A} = \mathcal{A}_t. \quad (34)$$

- 3) **Reward:** The design of the reward function should align with the objective of minimizing the completion time of data collection. The reward function should consist of both positive and negative rewards to provide appropriate incentives or penalties for the agent's actions. The positive reward component should encourage the agent to make progress towards completing the mission. This can be achieved by providing a reward when data is successfully collected and offloaded, indicating progress towards the completion of the task. On the other hand, the negative reward component should discourage delays or inefficient actions that could hinder the completion time. This can be achieved by imposing a penalty for delays in data collection. Specifically,

$$r(t) = -\delta + w_d [\Delta^S(t) + \Delta^A(t)], \quad (35)$$

where $\Delta^S(t)$ and $\Delta^A(t)$ are the amount of data offloaded in the backhaul link and collected in the access link, respectively, and w_d is a coefficient that balances the reward for data collection and the penalty for delay.

In the proposed dynamic SAGIN, the system state transitions from one time slot to another based on the actions taken in previous time slots. Specifically,

$$C(t) = \min \left\{ C, C(t-1) + R^S(t) \delta \sum_{i=1}^{|Z|} a_i(t) - \sum_{i=1}^{|Z|} R_i(t) \delta a_i(t) \right\}, \forall t \in \mathcal{T}, \quad (36)$$

$$D_i(t) = \max \{0, D_i(t-1) - \min [R_i(t-1) \delta w(t) \times a_i(t-1), C(t-1)]\}, \forall i \in \mathcal{I}, \forall t \in \mathcal{T}. \quad (37)$$

B. The Learning Rate

To achieve faster convergence while ensuring learning stability, we set the learning rate to be a relatively higher value in the initial stage of learning and decrease it in the later stage, as

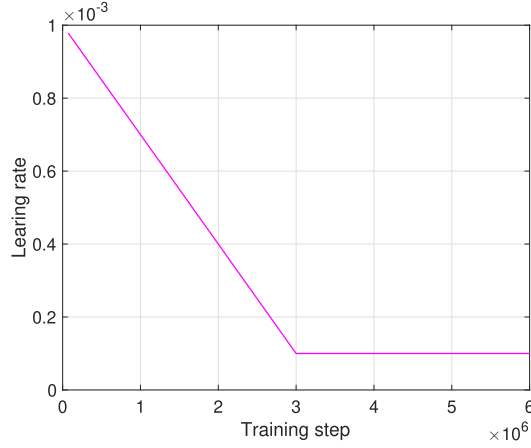


Fig. 4. Example of the learning rate.

follows,

$$l_r = \begin{cases} l_0 - \frac{(l_0 - l_1)}{m}t, & \text{if } t \leq 0.5T_s, \\ l_1, & \text{if } t > 0.5T_s, \end{cases} \quad (38)$$

where l_0 and l_1 represent the initial and later stage learning rates, respectively. m is the threshold that determines the transition point between the initial larger learning rate and the later smaller learning rate. T_s denotes the maximum allowed time step. Fig. 4 gives an example of the learning rate with $l_0 = 0.001$, $l_1 = 0.0001$, $m = 0.5$ and $T_s = 6 \times 10^6$. By starting with a larger learning rate in the beginning, the agent can converge faster towards a good solution. As the training progresses, the learning rate is reduced to a smaller value, slowing down the updates to the model's parameters. This adjustment helps to fine-tune the learned representations and make more precise adjustments to the parameters. Slowing down the learning rate in later stages of training can also prevent overshooting and oscillations, leading to a more stable convergence.

C. DRL-Based Algorithm

Next, we apply the Q-learning technique to tackle the reformulated MDP. However, due to the involvement of large state spaces in our problem, using a conventional Q-table would result in a significant increase in memory requirements. To overcome this challenge, we adopt a different approach by leveraging deep neural networks, as proposed in the work [17], to represent the relationship between the action and state space. By utilizing deep neural networks as function approximators, we can effectively handle the large state spaces and capture the complex interactions between actions and states. This allows us to approximate the Q-values and make efficient decisions in our Q-learning framework.

The Q network parameters are updated by minimizing the loss function:

$$\mathcal{L}(\theta) = (y_t - Q(s_t, a_t; \theta))^2, \quad (39)$$

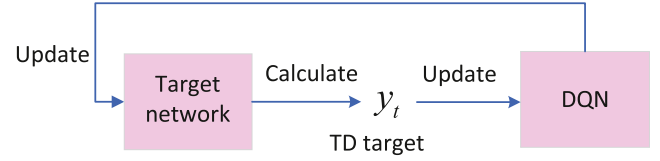


Fig. 5. DQN with target network.

where s_t and a_t are state and action at time step t , respectively. θ are the weights of the Q network. $Q(s_t, a_t; \theta)$ is the approximated action-value function at time step t , and

$$y_t = \begin{cases} r_t, & \text{if episode terminates at step } t+1, \\ r_t + \gamma \max_{a'} \hat{Q}(s_{t+1}, a'; \hat{\theta}), & \text{otherwise,} \end{cases} \quad (40)$$

where r_t is the reward at time step t , γ is the discount factor that specifies the weight of future rewards, and $\hat{\theta}$ is the vector/matrix representing weights of the target $\hat{Q}$ network. $\mathcal{L}(\theta)$ measures the discrepancy between the predicted Q-value $Q(s_t, a_t; \theta)$ and the target Q-value y_t . The objective of the optimization process is to improve the accuracy of the neural network in approximating the Q-values and ultimately enhance the performance of the Q-learning algorithm. Note that the target network, as shown in Fig. 5, is introduced to reduce the overestimation and avoid the error propagation, which are caused by the maximization operation when calculating the temporal difference (TD) target. The fixed parameters in the target network provide a consistent reference for calculating the TD target, reducing the potential for overestimation and error accumulation. The delayed updates of the target network ensure that the learning process has a more stable target to learn from, thus improving the stability and convergence of the Q-learning algorithm. The details of the DRL-based algorithm is shown in Algorithm 1. The computational complexity of the proposed algorithm is determined by the operations in each iteration. For a fully connected neural network with K hidden layers, the corresponding time complexity can be calculated as [17] $\mathcal{O}(n_I n_1 + \sum_{k=1}^{K-1} n_k n_{k+1} + n_K n_O)$. Here n_I , n_O and $n_k, \forall k \in \{1, 3, \dots, K\}$ denote the number of neurons in the input layer, the output layer and the k -th hidden layer, respectively. Consequently, the complexity of each iteration in Algorithm 1 can be expressed as $\mathcal{O}(|\mathcal{S}|(n_1) + \sum_{k=1}^{K-1} (n_k n_{k+1}) + n_K |\mathcal{A}|)$, where $|\mathcal{S}|$ and $|\mathcal{A}|$ stand for the cardinality of state space and action space, respectively.

V. NUMERICAL RESULTS

We consider a $3 \text{ km} \times 3 \text{ km}$ area with 5 IoRT devices located at (300, 300), (300, 2700), (1500, 1500), (2700, 300), and (2700, 2700), respectively. The flying height of the UAV H is set to 100 m. The initial and final point of the UAV is located at $Q = (0, 0)$. The data size of the IoRT devices are generated using a Gaussian random distribution with an average of 2 GB and a standard deviation of 100 MB. Unless specified otherwise, the simulation employs the parameters presented in Table I. For the DQN implementation, we utilize PyTorch. The DQN model comprises two hidden layers, each containing 128 neurons. The

Algorithm 1: DRL-Based Algorithm.

- 1: Initialize the Q network and target $\hat{Q}$ network with randomly values of θ and $\hat{\theta}$, respectively;
- 2: Initialize the replay buffer;
- 3: **for** $step = 1, T_s$ **do**
- 4: Obtain action a_t according to the ϵ greedy policy;
- 5: Take action a_t and obtain the reward r_t and the next state s_{t+1} ;
- 6: Store the tuple (s_t, a_t, r_t, s_{t+1}) in the replay buffer;
- 7: Randomly obtain N transitions from the replay buffer;
- 8: Obtain the value of the loss function, (39);
- 9: Take a gradient descent step with respect to the parameters of Q -network θ ;
- 10: Set $\hat{Q} = Q$ at every fixed number of steps;
- 11: **end for**

TABLE I
SIMULATION PARAMETERS

Parameters	Definitions	Values
γ_0	Pathloss at the reference distance	42 dB
σ^2	Noise power	-80 dBm
B	Total available bandwidth	10 MHz
P	Transmit power of IoRT devices	0.3 W
V^m	Max velocity of UAV	20 m/s
C	Cache capacity	1.5 GB
μ_D	Average data size	2 GB
σ_D	Deviation of data size	100 MB
δ	Duration of each time slot	0.5 s
L	Buffer size	200,000
γ	Discount factor	0.99
l_0	Initial learning rate	0.001
l_1	Final learning rate	0.0001

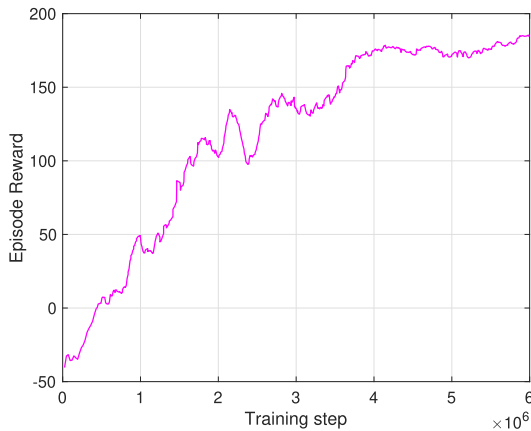


Fig. 6. Episode reward.

buffer size is set to 200,000, and the discount factor, which determines the weight of future rewards, is set to 0.99.

Figs. 6, 7 and 8 show the performance of our proposed DRL-based algorithm. As observed in Fig. 6, the reward value gradually increases as the number of training steps progresses. This upward trend signifies that the algorithm is learning to make improved decisions and accomplish its objectives more effectively. Fig. 7 shows that the training loss gradually decreases

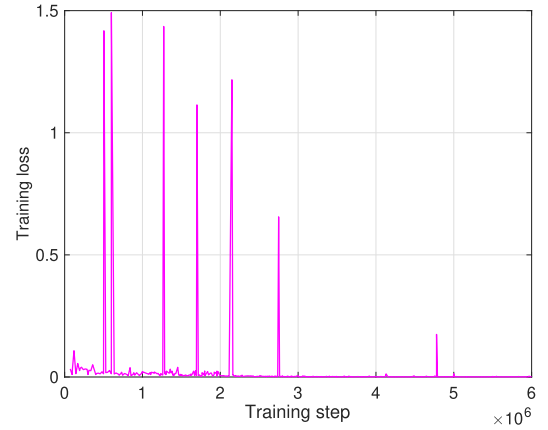


Fig. 7. Training loss.

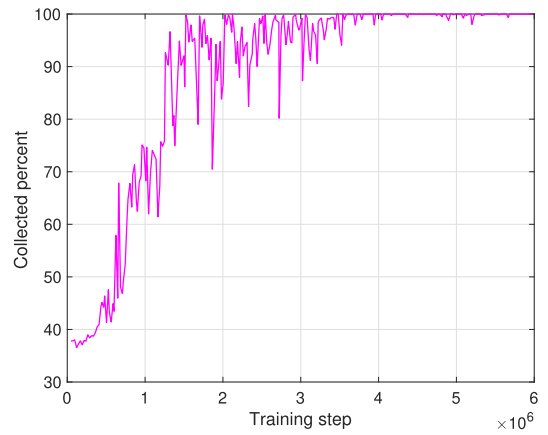


Fig. 8. Collected percent.

over the course of learning. The fluctuations in the training loss can be attributed to the randomness introduced by stochastic gradient descent (SGD) optimization or the variability present in the mini-batches. The decreasing training loss suggests that the model is converging towards a better representation of the problem and is becoming more accurate in predicting the desired outcomes. Fig. 8 provides validation of the effectiveness of the proposed algorithm by demonstrating the gradual increase in the collected percentage of data as the training steps progress. The result shows that the algorithm successfully achieves a collected data percentage of 100%, indicating that the mission can be completed using the DRL-based algorithm. Overall, the above three figures verify the convergence and effectiveness of our proposed algorithm.

Fig. 9 illustrates a snapshot of the UAV's trajectory obtained using the proposed algorithm and depicts the variation in the cached data size within the UAV. Note that the UAV starts by approaching the nearest IoRT device. In the meantime, the amount of cached data in the UAV continuously increases during this phase because the backhaul link capacity is not large enough to accommodate all the data collected in the access link and the excess data has to be stored in the UAV temporarily. However, as the backhaul link capacity surpasses that of the access link, the cached data size gradually decreases, as observed from steps

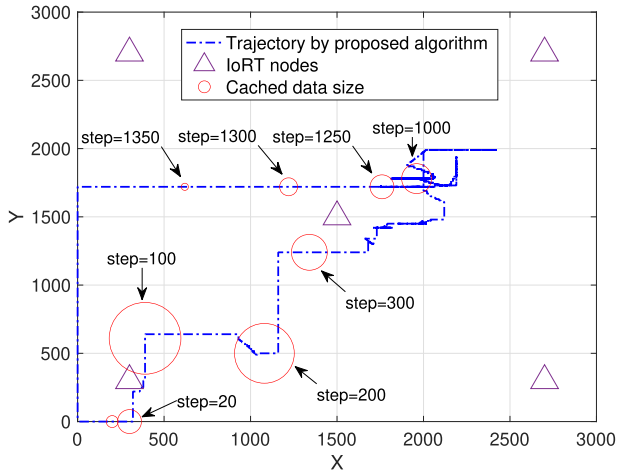


Fig. 9. Snapshot of UAV's trajectory obtained by the proposed algorithm.

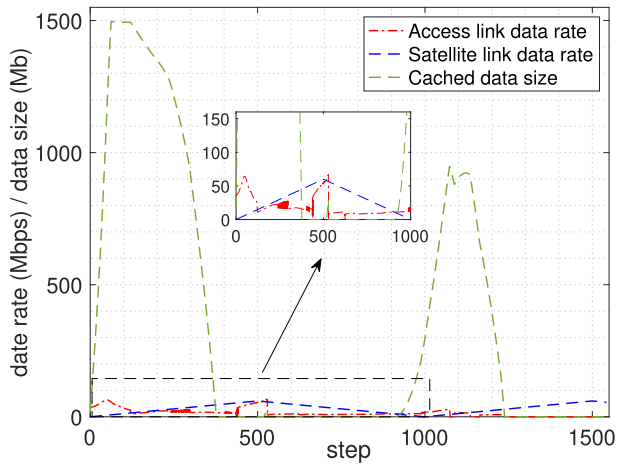


Fig. 10. Variation of cached data size.

100 to 300 and steps 1000 to 1350. The above statement can also be verified in Fig. 10, which shows how the cached data size changes as the backhaul link fluctuates. Note that we model the fluctuation of the backhaul link using a triangle wave with an amplitude of 60 Mbps and a period of 500 s (1000 steps).

Fig. 11 illustrates the trajectory of the UAV obtained by the proposed algorithm in two scenarios: even data distribution and uneven data distribution. In the even distribution scenario, all IoRT device have the same data size of 2 GB. However, in the uneven distribution scenario, the data size of the IoRT device located at the upper right (2700, 2700) is increased to 4 GB. It can be observed that the UAV approaches the IoRT device with the larger data size more closely in the uneven distribution scenario because the time saved in data collection by approaching the IoRT device with a larger data size outweighs the additional distance traveled by the UAV. This result demonstrates the adaptability of our proposed algorithm and its intelligent adjustment to minimize the completion time in different scenarios.

Figs. 12, 13 and 14 compare the performance of the proposed algorithm with two baseline algorithms, i.e., the circular trajectory algorithm and the proposed algorithm without cache. In the circular trajectory algorithm, the UAV collects data from each

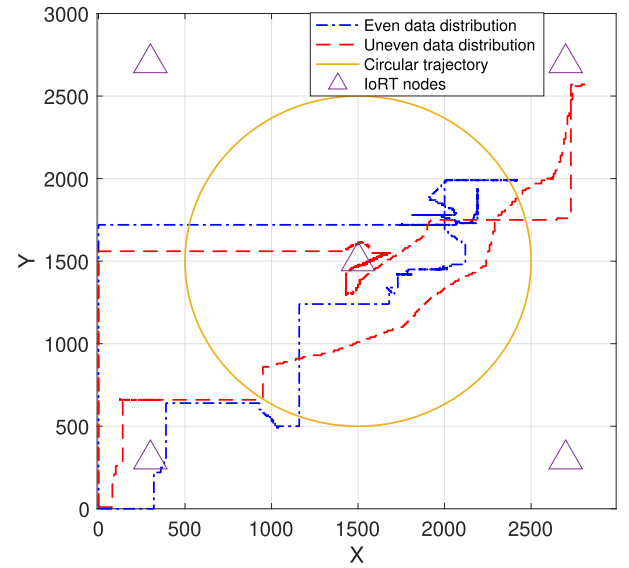


Fig. 11. UAV's trajectory with even and uneven IoRT data size.

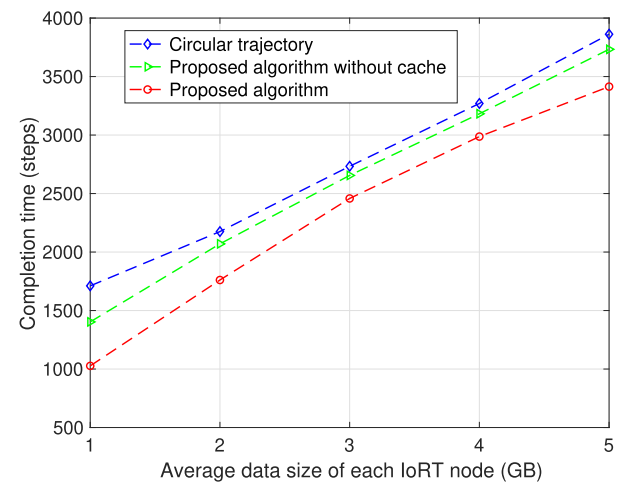


Fig. 12. Completion time vs average data size.

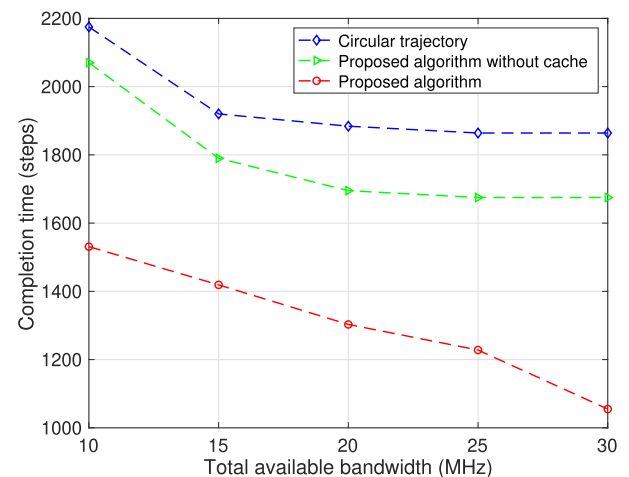


Fig. 13. Completion time vs total available bandwidth.

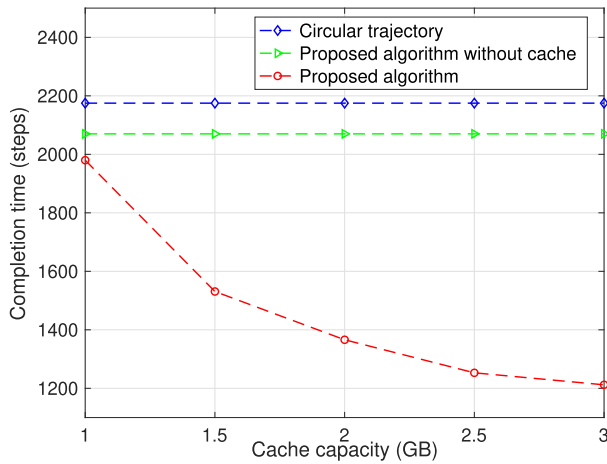


Fig. 14. Completion time vs cache capacity.

IoRT device at the nearest location and hovers around until the data collection is complete. The UAV is flying at the maximum allowed speed during the transition from one IoRT device to the next. In the proposed algorithm without cache, all the parameters are maintained the same as the proposed algorithm except that the UAV does not have a cache node. In Fig. 12, it can be observed that as the average data size of each IoRT device increases, the completion time of all three algorithms also increases. However, the proposed algorithm exhibits the shortest completion time compared to the two baseline algorithms. This is attributed to the adaptability of the proposed algorithm in adjusting the UAV's trajectory and utilizing the cache to store collected data during periods of low data rates in the backhaul link.

In Fig. 13, we can observe that the reduction in completion time for the two baseline algorithms is insignificant when the total available bandwidth exceeds 20 MHz, and even with no further reduction after $B \geq 25$ MHz. This is because increasing the available bandwidth only enhances the data rate in the access link, while the system throughput remains limited by the backhaul link as the UAV has no cache. On the other hand, the completion time of the proposed algorithm continues to decrease as the total available bandwidth increases, attributed to the cache node equipped on the UAV. The increased available bandwidth reduces the time required in the access link to temporarily store the collected data during periods of low data rates in the backhaul link. By effectively utilizing the cache, the proposed algorithm takes advantage of the increased bandwidth to further optimize the completion time.

Fig. 14 illustrates the impact of cache capacity on the completion time of data collection. Since the two baseline algorithms do not have a cache node on the UAV, their completion times remain unaffected by changes in the cache capacity. However, the completion time achieved by the proposed algorithm decreases as the cache capacity increases. This is because a larger cache capacity allows for more storage of collected data during low data rate periods in the backhaul link, potentially reducing the data collection time during such periods, thereby further reducing the overall completion time.

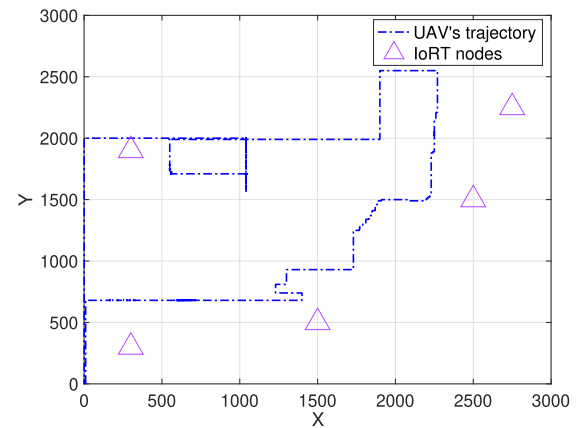


Fig. 15. UAV's trajectory with non-uniformly distributed IoRT devices.

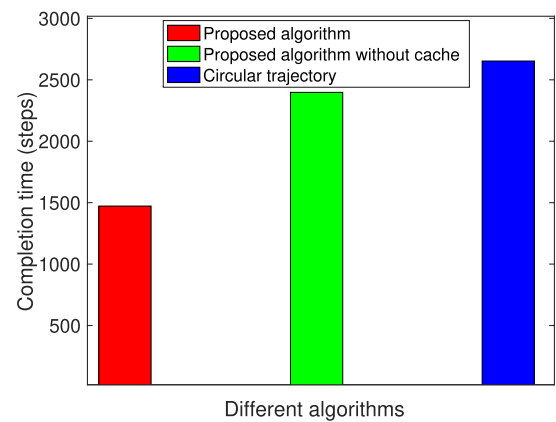


Fig. 16. Completion time with non-uniformly distributed IoRT devices.

Fig. 15 illustrates the UAV's trajectory in a scenario with non-uniformly distributed IoRT devices. In this case, the five IoRT devices are located at (300, 300), (1500, 500), (2500, 1500), (2750, 2250) and (300, 900), respectively. All other parameters remain consistent with the scenario where IoRT devices are uniformly distributed. The figure demonstrates that, using the proposed algorithm, the UAV can adaptively adjust its trajectory and approach the IoRT devices properly to potentially reduce the completion time. Furthermore, Fig. 16 confirms that the performance of the proposed algorithm remains superior to the other two benchmark algorithms, just as it does when IoRT devices are uniformly distributed. The reduced completion time is attributed to the cache node, which acts as a buffer between the access link and the backhaul link. During periods of low data rates in the satellite link, data can be collected and temporarily stored in the cache node without being constrained by the limitations of the satellite link. When the satellite link conditions improve, the cached data can be offloaded to the satellite. This strategy enables the full utilization of the satellite link's capacity during high data rates and helps circumvent the constraints imposed by the satellite link during periods of low data rates, resulting in less time taken by the proposed algorithm.

VI. CONCLUSION

In this article, we have considered the problem of minimizing the completion time for IoRT data collection in a LEO satellite-assisted and cache-enabled UAV network. Our approach involves designing the UAV's trajectory, the association scheme of IoRT devices, and the caching policy for collected data to achieve this objective. Unlike existing works, we have taken into account the fluctuation of the backhaul link capacity and proposed equipping the UAV with a cache node to mitigate the impact during low data rate periods. The formulated problem is non-convex, dynamic, and involves an infinite number of decision variables. To solve the formulated problem, we have reformulated it as an MDP and developed a DRL-based algorithm to find near-optimal solutions. We have conducted extensive numerical simulations to validate the effectiveness of the proposed algorithm under various scenarios. The simulation results have demonstrated that our proposed algorithm can adapt to different setups effectively. Moreover, the incorporation of a cache node on the UAV can effectively mitigate the influence of backhaul link capacity fluctuations, resulting in a reduction in completion time when compared to the two baseline algorithms. These findings highlight the benefits of our approach in optimizing IoRT data collection and improving overall system performance.

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